

Chemistry — Unit 5: Molecular Shape

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Quiz [A] — Bonding Sites, Lone Pairs, Shape Names, and 3-D Drawing

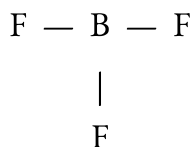
Name: _____ Block: _____ Date: _____

Section 1 — Identify Bonding Sites and Lone Pairs (5.1.1)

For each structural formula shown, count the bonding sites and lone pairs on the central atom.

1. The structural formula for BF_3 is shown below. Boron (B) is the central atom.

(4 pts)



- a. How many **bonding sites** does B have? _____
b. How many **lone pairs** does B have? _____

2. The structural formula for H_2O is shown below. Oxygen (O) is the central atom.

(6 pts)



- a. How many **bonding sites** does O have? _____
b. How many **lone pairs** does O have? _____
c. What is the **total number of electron pairs** (bonds + lone pairs) around O? _____

3. The structural formula for SO_2 is shown below. Sulfur (S) is the central atom.

(6 pts)



- a. How many **bonding sites** does S have? _____
b. How many **lone pairs** does S have? _____
c. SO_3 has 3 bonding sites and 0 lone pairs on S. How does this make SO_3 *different* from SO_2 ?

Section 2 — Name Molecular Shapes (5.1.2)

4. Fill in the blanks for each combination of bonds and lone pairs.

(4 pts)

- a. 2 bonding sites, 0 lone pairs → Shape: _____ Bond angle: _____°
b. 4 bonding sites, 0 lone pairs → Shape: _____ Bond angle: _____°

5. A molecule has a central atom with 3 bonding sites and 1 lone pair.

(6 pts)

- a. What is the name of the molecular shape? _____
- b. What is the approximate bond angle? _____ °
- c. A molecule with 3 bonding sites and 0 lone pairs has a bond angle of 120°. Why is this molecule's angle smaller?
- _____

6. Complete the table. For each molecule, fill in the bonding sites, lone pairs, and shape name.

(6 pts)

Molecule	# Bonding Sites on Central Atom	# Lone Pairs on Central Atom	Shape Name
CH ₄			
NH ₃			
H ₂ O			

Section 3 — Draw 3-D Structures Using Wedges and Dashes (5.1.3)

Reminder: Regular line = bond in the plane of the paper | Solid wedge (▲) = bond coming toward you | Dashed wedge (---) = bond going away from you.

7. Draw the 3-D structure of CH₄ (methane) using wedges and dashes.

(8 pts)

Label each bond as “wedge,” “dash,” or “regular.” Place an H at the end of each bond. (4 bonds total)

drawing area

Scoring: 1 pt each for correctly drawn wedge, dash, and 2 regular bonds (4 pts) + 1 pt each for correctly placed H atoms (4 pts)

8. CHALLENGE — Draw the 3-D structure of NH₃ (ammonia) using wedges and dashes.

(10 pts)

Include the lone pair in your drawing (label it “lone pair” or draw two dots). Label each bond type.

Then, in the space below your drawing, explain in one sentence why NH₃ is trigonal pyramidal and not tetrahedral.

drawing area

Explanation:

Scoring: 3 pts drawing (correct shape) + 3 pts labels + 1 pt lone pair shown + 3 pts explanation

Total: _____ / 50